

DSAI Project Milestone 6 Non-Technical Report

About the Project

This project is about making it faster and cheaper for computer systems to answer questions written in Indian languages. When a computer answers a question, it is usually given a large amount of background text (called a "prompt") that contains the information needed to work out the answer. The longer this background text is, the more time, computing power, and cost it takes for the system to process it.

The goal of this project is to build a tool that automatically shortens this background text by keeping only the most important words and removing the rest, while still keeping enough information for the system to answer the question correctly. This process is called "prompt compression."

Shortening the text given to an AI system, without losing important meaning, has several real-world benefits:

- It reduces the computing cost and time needed to process each question.
- It allows AI systems to handle more questions using the same resources.
- It is especially useful for Indian languages, which are often under-supported compared to English in AI research.

In short, this project aims to make question-answering systems for Indian languages more efficient without sacrificing accuracy.

3. How the Tool Works

Rather than asking the system to rewrite the background text from scratch, the project takes a simpler and more reliable approach: it looks at each individual word in the text and decides whether that word should be "kept" or "removed." A trained AI model was taught to make this keep-or-remove decision for every word, based on examples where a shortened version of the text was already known to work well.

To teach the model, the project used a well-known multilingual AI language model as a starting point, and then trained it specifically on Indian-language question-and-answer examples so it could learn which words matter most for answering a given question.

4. Data Used for Testing

The data used in this project came from a large collection of Indian-language question-and-answer examples, split into three separate groups so that the results could be trusted:

- Training set (1,610 examples) – used to teach the model.

- Validation set (202 examples) – used to fine-tune and compare different versions of the model.
- Test set (203 examples) – kept completely separate and used only at the end, to check how well the final model performs on data it had never seen before.

5. How the Model Was Trained and Checked?

The team tried several different training settings to find the combination that produced the best results. All the training and testing was carried out on cloud-based computers with graphics processing hardware suited for AI work, using standard, widely trusted AI software tools. A fixed, repeatable process was used throughout, so the results could be reliably reproduced. To judge how well the model was performing, four standard performance measures were used: Accuracy (how often the model was correct overall), Precision (how often the words it chose to keep were actually important), Recall (how many of the truly important words it successfully identified), and F1-score, which balances Precision and Recall together. F1-score was treated as the most important measure, since the project cares equally about not missing important words and not keeping unnecessary ones.

6. Results

After testing different training settings, the best-performing version of the model was selected and trained further, then evaluated on the separate test data that it had never seen before. The final results were as follows:

Measure	Score	What it means?
Accuracy	0.73	The model made the correct keep/remove decision about 73% of the time.
Precision	0.53	Of the words the model chose to keep, about 53% were genuinely important.
Recall	0.67	The model successfully identified about 67% of all the truly important words.
F1-score	0.59	The overall balance of correctness across both precision and recall.

Two additional checks were used to understand the model's overall ability to tell important words apart from unimportant ones:

- A measure of how well the model can distinguish between important and unimportant words across many possible decision points came out to about 0.83 out of a possible 1.0, which indicates fairly strong ability to tell the two apart.
- A second measure, which looks specifically at how well the model balances precision and recall when identifying important words, came out to about 0.60, suggesting a reasonable but not perfect balance.
- Overall, these results show that the tool performs reasonably well: it correctly identifies most important words most of the time, though there is still noticeable room for improvement, particularly in reducing incorrect "keep" decisions.

7. What the Tool Does Well

- It correctly handles the majority of words across most of the text samples that were tested.
- Its performance does not get noticeably worse as the input text gets longer, which means it can be used reliably on both short and long questions.
- It shows a strong overall ability to distinguish important words from unimportant ones, as reflected in its discrimination score of about 0.83.

8. Where the Tool Still Struggles

Like any AI system, this tool is not perfect and has some known limitations:

- A small number of specific words are consistently misclassified almost every time they appear. These tend to be less common words in the language, which the model has not seen often enough during training to learn confidently.
- Mistakes are slightly more likely to happen toward the later part of a longer piece of text, suggesting the model finds it a little harder to judge importance once it has processed a lot of preceding context.
- The balance between correctly keeping important words and avoiding unnecessary ones (the F1-score of 0.59) shows that, while the tool is useful, it is not yet at a level where it can be relied upon without any review, especially for sensitive or high-stakes uses.

Interestingly, the project also found that simply achieving a lower training "error" during model training did not always translate into better real-world performance, which highlights why multiple performance measures — rather than just one — are needed to properly judge an AI model.

9. Overall Conclusion

This project successfully built and tested a working tool that shortens question-answering prompts in Indian languages by automatically identifying and keeping only the most important words. Through testing several training approaches, the team identified a well-performing configuration and confirmed its effectiveness on completely unseen data, achieving solid, if not yet perfect, results.

The tool shows clear promise for making Indian-language question-answering systems faster and more efficient, while also highlighting specific areas — such as handling rare words and longer texts — that future work can focus on to further improve reliability.